Lecture 9: Evolution, Emotion, and Reason: Love

Introduction to Psychology · Yale University, Spring 2007 · Guest lecture by Professor Peter Salovey

Study guide based on the lecture transcript. Source: https://oyc.yale.edu/psychology/psyc-110/lecture-9 (CC BY-NC-SA 3.0).

Before you start

In this Valentine's Day guest lecture, Peter Salovey, a social psychologist and then Dean of Yale College, talks about love. He first defines love with three parts. Then he describes experiments on what makes people attracted to each other. Many results are surprising, and some have serious real-life implications.

Learning goals

After this lecture you should be able to:

- Describe Sternberg's three components of love and the types of love they form.
- Explain the "big three" factors in attraction: proximity, similarity and familiarity.
- Describe the Pratfall Effect, the role of physical attractiveness, and the gain-loss effect.
- Explain misattribution of arousal and give examples of experiments that show it.
- Explain why random assignment is needed in a true experiment.

Key ideas

1. Introduction to Dr. Peter Salovey

Professor Bloom introduces Peter Salovey, then Dean of Yale College. He is a social psychologist who studies health psychology, and the founder and developer of the idea of emotional intelligence. He was also a famous Introduction to Psychology teacher at Yale.

2. Defining Love and Its Permutations

- Two warnings: some experiments from the 1950s to the early 1970s would not be ethically acceptable today. Also, most studies used men as participants and women as targets, so we should ask if the results apply to other relationships.
- Robert Sternberg's theory: love has three components. **Intimacy** is closeness and bonding, like sharing secrets. **Passion** is physical attraction. **Decision/commitment** means calling it love and wanting to keep the relationship.
- The combinations: none = non-love; intimacy only = liking (a casual friendship); passion only = infatuation (love at first sight); commitment only = empty love; intimacy + passion = romantic love (Romeo and Juliet at first); intimacy + commitment = companionate love (a best friend); passion + commitment = fatuous love (a whirlwind "Hollywood romance"); all three = consummate love.
- Empty love is often the last stage of a long relationship that went bad. In arranged marriages, it is often the first stage. Such marriages do not end in divorce more often than love marriages, but

there is a confound (another factor that could explain the result): they happen more in societies where divorce is very costly.

3. The Social Psychology of Love and Attraction

- Most research is about attraction. Salovey groups seven factors into "the big three" and "the more interesting four."
- **The big three** ("all other things being equal"): **proximity** (people who live closer, counted in Manhattan city blocks or in feet between dorm doors, more often form relationships); **similarity** ("birds of a feather flock together"; opposites do not really attract); **familiarity** (we usually fall for someone we already know, not a stranger "across a crowded room"). We even like words in an unknown language more if we saw them before, without remembering it.
- **Competence and the Pratfall Effect:** a very competent person who makes a small mistake is liked even more. In Elliot Aronson's experiment, students heard a taped interview with a candidate for the quiz show "College Bowl": a nearly perfect one (92% correct) or a mediocre one (30%). When the candidate spilled coffee on his new suit, the nearly perfect one was rated even more attractive, and the mediocre one even less. As a new graduate student, Salovey spilled coffee on Judy Rodin's desk; she told him the effect "only works if I think you're competent first."
- **Physical attractiveness:** students say looks are not important. But after blind dates, looks (not warmth, intelligence or humor) predicted a second date. In Aronson's "Frizzy Wig" experiment, a confederate (a woman working for the experimenter) gave men a good or bad evaluation of their personality. Her feedback mattered a lot when she looked attractive, but little when she wore a frizzy wig and unattractive clothes.
- **Gain and loss:** we are more sensitive to change than to steady states. In the "Gain Effect," we are most attracted to people whose liking for us grows over time. And the people who hurt us most are those whose love for us fades.

4. Misattribution for the Causes of Arousal

- **Misattribution of arousal:** your body is aroused (fast heartbeat, sweaty palms) for one reason, but you believe the cause is something else, such as love.
- Thought experiment: a friend orders decaf, but the barista gives her caffeinated espresso by mistake. Her heart beats faster, and she thinks she is falling in love with the person beside her.
- The "Rickety Bridge" study (University of British Columbia): an attractive woman met men who had just crossed a high, swaying rope bridge or a low, solid bridge. Men from the scary bridge wrote sexier stories and were more likely to call her later. But it was not a true experiment: without random assignment (putting people into groups by chance), adventurous men may have chosen the scary bridge.
- Lab experiments fixed this. People waiting for a "painful shock experiment," or who had run on a treadmill for ten minutes, rated the experimenter as more attractive.
- In another study, men looked at magazine photographs of nude women while hearing what they thought was their own heartbeat. It was a tape that the experimenter sped up at random. The men preferred the photographs shown with a faster heartbeat.

- A serious implication: in abusive relationships, people usually stay for economic reasons, but some may also misread anger, aggression or their own fear as love.

5. Question and Answer

The more obvious the real cause of arousal, the weaker the effect. The big three probably matter for keeping relationships too, but research mostly studies first attraction.

Key vocabulary

Term	Meaning in simple English	Example from the lecture
intimacy	closeness and bonding; sharing secrets	telling one person things you tell no one else
passion	physical attraction	infatuation, or "love at first sight"
commitment	deciding to call it love and to keep the relationship	empty love has only commitment
companionate love	intimacy and commitment without passion	a best friend
proximity	being physically near	the number of feet between dorm room doors
familiarity	already knowing someone or something	liking words you saw before
Pratfall Effect	a competent person becomes more liked after a small mistake	spilling coffee on a new suit
confederate	a person who works for the experimenter and plays a role	the woman in the frizzy wig
confound	another factor that could explain a result	divorce is very costly in some societies
Gain Effect	strong attraction to someone whose liking for us grows	regard that starts low and rises
arousal	the body's excited state: fast heartbeat, sweaty palms	after caffeine or a scary bridge
misattribution	believing the wrong cause for something	thinking caffeine effects are love
random assignment	putting participants into groups by chance	missing in the Rickety Bridge study

Check your understanding

1. What are the three components of love in Sternberg's theory?
2. What does Sternberg call a relationship with commitment but no intimacy or passion? When does it often happen?
3. What are the "big three" of attraction?

4. In Aronson's coffee-spilling experiment, how did the blunder change the ratings of the two candidates?
5. What do college students say about looks, and what do the studies show?
6. Explain the Gain Effect in your own words.
7. What was the flaw in the Rickety Bridge study? How did the lab studies fix it?
8. After a scary film, a man's heart is beating fast, and he suddenly feels he loves the woman beside him. How would Salovey explain this?

Answer guide

1. Intimacy, passion, and decision/commitment.
2. Empty love. It is often the last stage of a long relationship that went bad, or the first stage of an arranged marriage.
3. Proximity, similarity and familiarity.
4. The nearly perfect candidate was rated even more attractive after the blunder. The mediocre candidate was rated even less attractive.
5. They say looks are not important, but looks predicted who went on a second date better than warmth, intelligence or humor.
6. We are most attracted to people whose liking for us increases over time, more than to people who always liked us.
7. There was no random assignment, so different kinds of men may have chosen each bridge. Lab studies put people into groups by chance (shock instructions or a treadmill).
8. Misattribution of arousal: his body is aroused by fear, but he believes the cause is attraction to the woman.

English practice

Useful phrases from the lecture

- "All other things being equal, ..." - shows a general tendency when other factors are the same.
- "What's interesting is..." - points to the surprising part of an idea.
- "Another way of saying it is..." - repeats an idea in different words.
- "If this were the end of the story though, it would be a kind of boring story." - shows that a surprising result is coming next.
- "I'm not saying this is fair, ... but empirically..." - separates a research result from your own opinion.

Speaking task (with your AI tutor)

- Explain Sternberg's theory of love in 4-6 sentences. Use: *intimacy*, *passion*, *commitment*.
- Explain the Rickety Bridge study and its flaw in 4-6 sentences. Use: *arousal*, *misattribution*, *random assignment*.

Writing task

- "Opposites attract." Is this true? Write one paragraph (80-120 words) that answers using evidence from the lecture about similarity, proximity and familiarity.

Study with your tutor

Things to say to your tutor in Live conversation:

- "Let's review Lecture 9. Ask me questions about Sternberg's types of love, one at a time."

- "Quiz me on the key vocabulary of Lecture 9."
- "I will explain the Pratfall Effect in my own words. Check my understanding and my English."
- "Describe a situation, and I will say whether it shows misattribution of arousal."

Readings from the original course

- Gray, Peter. Psychology (5th edition), pp. 456-458

My notes